Supplementary Material:
Cross-Age Face Verification from Mined Same-Person Social-Media Pairs

Anonymous submission

This appendix reports the controls, audits, full operating-point tables, and the data/model card deferred from the main paper for space. Numbering here is internal to this document; references to the main paper are by name. All experiments use the curated data and the same weak-backbone attribution protocol described in the main paper.

1 Extended Related-Work Positioning

Table 1 positions our naturally-supervised source against the discriminative, contrastive and synthetic-aging lines of prior work. The common thread is that prior methods depend on *manual* identity and/or age labels or on synthesized cross-age samples, whereas we mine the structure "one post = one person at different ages" as a label-free positive-pair generator.

2 Data and Audit Details

Age extraction. Across all 33,697 captioned multi-photo posts, the LLM and regex agree on 72.8% (Table 2); the LLM fills a further 10.8% the regex missed and is favored in the bulk of the 13.8% mismatches (it avoids reading calendar years or year-gaps as ages).

Group integrity. Person-clustering had already separated most multi-person posts; an LLM group-integrity pass (Table 3) flags only 421 person-groups as noisy (multi-person/meme).

Dedup-threshold robustness. The -52% positive-count reduction from near-duplicate removal is not an artifact of the 0.97 cutoff: it is stable across 0.93–0.97 and degrades only at the very strict 0.99 (Table 4).

Small manual validation. One annotator spot-checked the automatic pipeline (Table 5). With a single annotator and no inter-annotator κ this *supports* but does not *certify* the supervision; a large multi-annotator audit remains beyond our resources, so residual label noise is acknowledged as a limitation. Samples are drawn from the curated corpus and were not used to tune thresholds or filtering rules.

3 Objective Comparison (Full)

Figure 1 plots the objective comparison summarized in the main paper: the contrastive, ArcFace and CosFace objectives coincide on cross-age transfer (within ± 0.006 on FG-NET large-gap), a noise-robust sub-center ArcFace adds nothing on the curated pairs, and only SphereFace—hardest to optimize on 2–3-shot identities without λ-annealing—trails. ArcFace/CosFace additionally forget slightly less on the easy benchmarks, a practical recommendation.

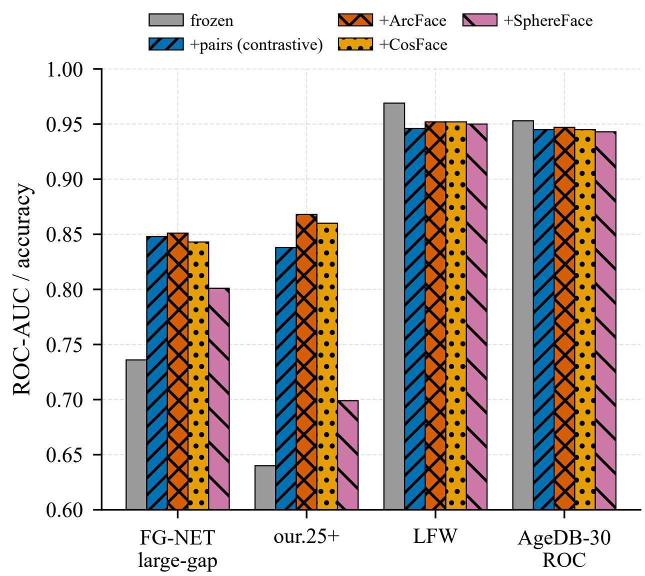

Figure 1: Objective comparison: contrastive, ArcFace and CosFace coincide on cross-age transfer; SphereFace trails. ArcFace/CosFace forget slightly less on easy benchmarks.

4 Headroom and Data-Scaling Tables

Table 6 gives the numeric headroom curve (the gain decreases monotonically with frozen strength, with a clean AdaFace IR-50→IR-101 depth control), and Table 7 the data-scaling law (the gain saturates by $\approx$10% of identities).

5 Mechanism: Age-Leakage Probe and Disentanglement

A linear probe (Table 8) shows apparent age stays well decodable from the identity embedding for *all* models ($\approx$0.63–0.65 balanced accuracy $\gg$ 0.25 chance) and changes only slightly, while identity AUC rises sharply—the method learns an invariant *metric* (the cosine stops relying on the age axis), not

Work	Supervision	Mechanism	Our difference
OE-CNN (2018) (Wang et al. 2018)	labeled id+age	orthogonal feature decomposition	no manual id/age labels, not decomposition
DAL (2019) (Wang et al. 2019)	labeled id+age	adversarial id/age factorization	mined pairs + simpler objective
MTLFace (2021) (Huang, Zhang, and Shan 2021)	labeled id+age	recognition + age synthesis (multi-task)	weaker pipeline; novel supervision origin
CACon (2023) (Wang, Sanchez, and Li 2024)	semi-sup. + synthetic	contrastive on synthesized cross-age	real mined pairs > synthetic aging
Synthetic Ageing (2024) (Yao et al. 2024)	synthetic	train on aged images	we make *real* mining the thesis
OrdCon (2025) (Wang, Sanchez, and Li 2025)	age-ordered contrastive	order-enhanced features	mine natural pairs, no explicit age order

Table 1: Positioning vs. prior work.

Category	n	Share
Agree (same ages or both empty)	24,531	72.8%
LLM filled (regex missed)	3,656	10.8%
Age mismatch	4,658	13.8%
LLM missed (regex found)	852	2.5%

Table 2: Age-extraction audit: LLM vs. regex (33,697 captioned multi-photo posts).

Category	n	Share
Single (one person across ages)	31,275	92.8%
Multi-person (different people)	2,034	6.0%
Unknown	336	1.0%
Meme / collage	52	0.2%

Table 3: LLM group-integrity classification (33,697 posts).

an age-free representation. Explicit adversarial disentanglement (gradient reversal + age head, MTLFace-style) reduces leakage no further and adds only a small cross-age gain over +pairs (+0.006 FG-NET large-gap, +0.015 our.25+); its absolute large-gap (≈0.853) matches the ArcFace objective (0.851), reinforcing that the data—not the objective or the disentangling add-on—sets the cross-age ceiling.

6 Apparent-Demographic Audit (Full)

Stratifying by *apparent* gender/age of the pair anchor (estimated, not self-identified), no group degrades and every gain is significant at the 95% paired-bootstrap level (Table 9, Fig. 2). The largest gain is for the youngest group 0–17 (+0.130), where the frozen baseline is weakest. We avoid the word "fair" as a solved property: the source is apparent-female-skewed (76.7%), attributes are apparent (not ethnicity

Dedup cos	Near-dup faces	Positives	Reduction
0.93	8,653	31,304	−52.5%
0.95	8,564	31,611	−52.0%
0.97 (used)	8,304	32,360	−50.9%
0.99	6,274	38,565	−41.5%

Table 4: Dedup-threshold sensitivity (near-duplicate removal alone, on pre-curation groups; the group-integrity prune removes a further ≈500 to reach the headline 31,865).

Component	n	Human-checked result
Age extraction	100	LLM 89%, regex 71% exact
Group integrity	100	single/multi acc. 0.96
Positive pairs	200	same-person prec. 0.96
Cluster merges	100	merge prec. 0.95
Near-dup removals	100	duplicate prec. 0.93

Table 5: Small manual validation (one annotator; a sanity check, *not* a certified audit; stratified/random/near-threshold sampling). The 100-post counts are illustrative and to be confirmed with the final annotation.

or legal categories), and 45+ is sparse (n=153).

At a fixed operating point (per-model global threshold at 1% FMR), +pairs lowers the false-non-match rate in *every* apparent stratum (0–17 $0.852 \to 0.656$, apparent-female $0.674 \to 0.509$, apparent-male $0.743 \to 0.534$), with per-stratum logistic calibration slopes uniformly positive (9.0–11.7); the only exception is the small 45+ stratum, whose FMR scatters at the global threshold (sampling noise). The child↔adult gain is corroborated on an independent FG-NET subset (one photo under 13, the other over 25; 195 positives): +pairs raises ROC-AUC $0.684\,[0.63, 0.74] \to$

Backbone (strength)	Frozen	+pairs	Δ large-gap	Δ our.25+
FaceNet (weak)	0.736	0.848	**+0.112**	+0.198
ArcFace r50 (weak, diff. arch.)	0.764	0.805	+0.041	+0.133
AdaFace IR-50 (strong)	0.917	0.924	+0.007	+0.004
ArcFace r100 (strong)	0.954	0.915	−0.039	−0.002
AdaFace IR-101 (strong)	0.957	0.931	−0.026	−0.006

Table 6: Headroom: FG-NET large-gap by backbone strength (curated, lr 3e-5). The gain is specific to weak/medium backbones with cross-age headroom.

Train fraction	# identities	FG-NET large-gap	our.25+
frozen	0	0.736	0.640
0.10	≈1,500	0.846 ± 0.002	0.816 ± 0.003
0.25	≈3,750	0.852 ± 0.000	0.842 ± 0.002
0.50	≈7,499	0.856 ± 0.005	0.845 ± 0.002
1.00	≈14,998	0.848 ± 0.001	0.835 ± 0.003

Table 7: Data-scaling law (FaceNet, curated; val/test fixed; mean ± std over three seeds).

Model	Age probe ↓	Identity AUC ↑
frozen	0.651 ± 0.019	0.836
+pairs	0.629 ± 0.012	0.916
+pairs+disent.	0.632 ± 0.017	0.918

Table 8: Age-leakage linear probe (curated; chance = 0.25). "Age probe" is the balanced accuracy of a linear age classifier read off the identity embedding (lower is better).

$0.813\,[0.77, 0.86]$ (+0.129, non-overlapping 95% CIs).

7 Cross-Source and Cross-Platform Transfer

Table 10 reports the VK-trained model evaluated on the independent non-VK Reddit then/now set, and Table 11 the bidirectional transfer (a backbone trained on *only* a small 220-person Reddit source also helps on VK).

8 Calibration and Hard-Negative Mining

Calibration. Raw cosine (Platt) is mis-calibrated across gap bins—over-confident at 15–25 years (ECE 0.052); a $P(\text{same} \mid \text{cosine}, \text{age-gap})$ calibrator fixes every bin (15–25 → 0.017) and improves the overall Brier score (0.035 → 0.029), yielding a usable, age-aware same-identity probability.

Hard-negative mining. Adding the top-5 most-similar other-person faces per anchor (in a hard cosine band, excluding the near-duplicate/uncertain range) further sharpens large-gap discrimination (Table 12): FG-NET large-gap rises to 0.868 ± 0.004 over three seeds, but easy benchmarks forget *more* and FG-NET overall ROC drops. Hard negatives thus trade general-benchmark accuracy for large-gap separation—useful for a cross-age retrieval setting, not a general verifier.

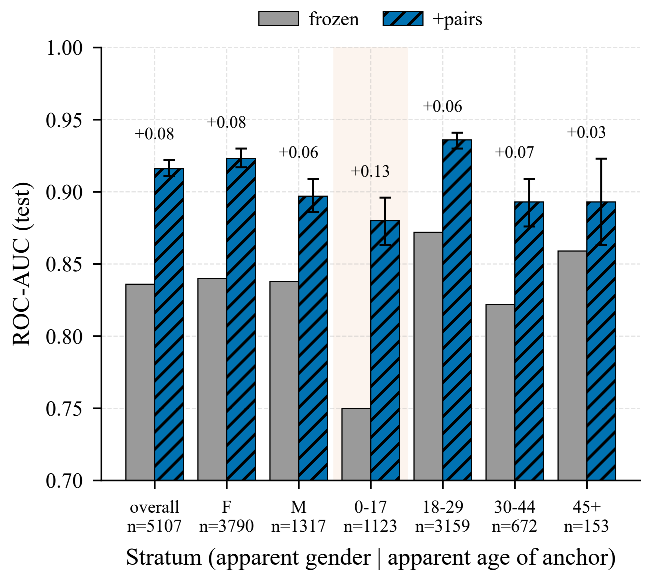

Figure 2: Apparent-demographic audit: frozen vs. +pairs ROC-AUC across apparent gender/age strata. Every stratum improves; the largest gain is for the youngest (0–17) group. Error bars are 95% paired-bootstrap CIs of the gain.

9 Claims vs. Evidence

Table 13 maps each substantive claim to its evidence and scope.

10 Model and Data Card

Intended use. Research on consent-based cross-age identity matching; the model outputs candidates for human review, never an automatic decision. **Out of scope:** mass identification, surveillance, de-anonymization, any processing of minors' data without a legal basis. **Training data.** Two public VK "then/now" communities (one platform, Russian-language context); apparent-female-skewed (76.7%); sparse at apparent age 45+; weak same-person supervision from same-post co-occurrence with LLM-parsed caption ages as a secondary cue. **Evaluation data.** LFW, AgeDB-30, CALFW, FG-NET (and its large-gap and child↔adult subsets), an internal curated cross-age test, and an independent non-VK Reddit then/now set. **Metrics.** ROC-AUC (threshold-free; primary), EER, TAR@FAR, and per-stratum calibration. **Ethical considerations.** Sensitive biometric data; no raw faces/posts released; weights/embeddings under controlled

Stratum	n_{pos}	Frozen	+pairs	Gain	95% CI
overall	5,107	0.836	0.916	+0.080	[+0.075, +0.086]
apparent F	3,790	0.840	0.923	+0.084	[+0.077, +0.090]
apparent M	1,317	0.838	0.897	+0.059	[+0.048, +0.071]
age 0–17	1,123	0.750	0.880	+**0.130**	[+0.113, +0.146]
age 18–29	3,159	0.872	0.936	+0.064	[+0.058, +0.069]
age 30–44	672	0.822	0.893	+0.070	[+0.054, +0.087]
age 45+	153	0.859	0.893	+0.034	[+0.004, +0.064]

Table 9: Apparent gender/age strata (FaceNet, curated; 95% paired-bootstrap CI of the gain). No stratum degrades.

Reddit (non-VK) test metric	Frozen	+pairs (VK-trained)
Overall ROC-AUC [95% CI]	0.718 [0.70, 0.74]	**0.749 [0.73, 0.77]**
EER	0.345	**0.328**
TAR@FAR=1%	0.271	**0.279**
Test pairs (pos / neg)	1,575 / 1,575	

Table 10: Cross-platform transfer: a VK-trained model on an independent Reddit (non-VK) then/now test set (396 posts → 1,575 positive pairs after the multi-person-collage filter).

Train	Test	Frozen	Tuned	Δ
VK	Reddit (overall)	0.718	0.749	+0.031
Reddit	FG-NET large-gap	0.736	0.783	+0.047
Reddit	VK internal 25+	0.640	0.659	+0.019

Table 11: Cross-source transfer, both directions. Reddit→VK is a small 220-person pilot supporting source non-specificity, not a headline claim.

Metric	Frozen	+pairs	+pairs+hard-neg
FG-NET large-gap	0.736	0.848	**0.868 ± 0.004**
FG-NET ROC	0.896	0.923	0.912 ± 0.003
LFW accuracy	0.969	0.950	0.925 ± 0.005
AgeDB-30 ROC	0.953	0.946	0.911 ± 0.013
CALFW ROC	0.948	0.949	0.898 ± 0.015

Table 12: Hard-negative mining (FaceNet, curated; external benchmarks). +hard-neg is mean ± std over three seeds.

access (DUA); minors withheld absent a legal framework; DPIA and datasheet (Gebru et al. 2021) accompany the release. **Caveats.** Targeted large-gap gain only (low-FAR operating points degrade); concentrates in weak/medium backbones; apparent (not self-identified) attributes; audited but not certified supervision.

11 Reproducibility Details

Pipeline. *Usable-face* filtering (detection score, minimum size, Laplacian blur, single target); detection/alignment with InsightFace `buffalo_l` RetinaFace + 5-point ArcFace `norm_crop` (112 px; FaceNet input 160 px); ArcFace `w600k_r50` embeddings for clustering (union-find, cosine $\geq$ 0.85) and dedup (cosine $\geq$ 0.97); an LLM with versioned, cached prompts for age and group-integrity parsing; a by-person 70/15/15 split (seed 42, balanced negatives, plus an age-matched negative variant). **Training.** Margin-contrastive head fine-tuning (lr 3e-5 weak / 1e-5 strong; early stop on validation AUC; seeds 42/1/2); margin identity-classification objectives (ArcFace/CosFace/SphereFace, and sub-center ArcFace with K=3) for the objective comparison; pure-NumPy ROC-AUC/EER/TAR@FAR. **Compute.** The entire study runs on a single laptop: one NVIDIA RTX 3060 Laptop GPU (6 GB), an AMD Ryzen 7 5800H (8C/16T) and 16 GB RAM under Windows 11; Python 3.12, PyTorch 2.12 (CUDA 13.2), InsightFace 1.0 / ONNX Runtime 1.26, scikit-learn 1.9, Matplotlib 3.11. **Availability.** Code, configurations, versioned LLM prompts, frozen splits (by hash), and the de-identified benchmark protocol are released; raw face images and posts are withheld, so data-mining replication is partial (we release cached LLM outputs for deterministic replay). A filled reproducibility checklist accompanies the submission.

Claim	Evidence	Scope / caveat
Mined pairs improve large-gap cross-age verification	FG-NET large-gap $0.736 \rightarrow 0.848$, non-overlapping CIs, DeLong $p < 10^{-19}$	weak/medium backbones
Data source $>$ objective choice	contrastive/ArcFace/CosFace coincide (0.848/0.851/0.843); sub-center adds nothing	among objectives tested
Real pairs $>$ synthetic aging	real 0.848 vs. gap-matched FRAN 0.759; native-res control	this protocol
Mechanism is age-shortcut removal	age-matched negatives collapse frozen to $\approx$0.52; $+$pairs recovers to 0.838	internal 25+ test
Source generalizes beyond platform	VK$\rightarrow$Reddit $+0.031$; child$\leftrightarrow$adult FG-NET $+0.129$	not platform-agnostic
Targeted, not general gain	easy-benchmark low-FAR degrades (LFW TAR@0.1% $0.858 \rightarrow 0.575$)	scope to large-gap retrieval
$\approx$half of raw positives were near-duplicates	-52% positive count, stable across dedup 0.93–0.97	curation finding

Table 13: Claims-vs-evidence map.

References

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